

Assignment 2

Multi-Head Attention & Representation Analysis

Strict Protocol Edition · Towards Research-Level Competence

Assignment 2

■ GLOBAL RULES

- Form a hypothesis BEFORE every experiment.
- Explain your reasoning before asking questions.
- After using external help, re-implement from memory.
- Every experiment must include: Hypothesis → Result → Interpretation.

Problem Statement

Extend your Assignment 1 single-head model into multi-head attention and rigorously study what each head learns. The task changes: each output position must emit the second-largest value in the input sequence.

- Input: integer sequences sampled from vocab size 16
- Task: predict the second-largest value at every output position
- Embedding dim ≤ 32 , use 2-4 heads, no LayerNorm, no residuals
- Loss: MSE first, then swap to cross-entropy and compare
- Minimum 5,000 synthetic training examples (you generate the dataset)

Implementation Requirements

1. Implement multi-head attention by splitting Q/K/V projections across heads
2. Concatenate head outputs and project through W_O
3. Each head must have its own independent Q, K, V weight matrices
4. Train on your self-generated synthetic dataset
5. Implement both MSE and cross-entropy loss; train separate runs with each
6. Do NOT use any framework attention modules -- implement every matmul yourself

Instrumentation Requirements

1. Log each head's attention matrix separately at steps 0, 100, 500, and final
2. Compute and plot the effective rank of each head's attention matrix over training
3. Track gradient norms per head -- measure whether heads learn at different rates
4. Visualize head specialization: does one head attend locally, another globally?
5. Plot W_Q , W_K , W_V spectral norms for each head over training

Failure Analysis (Required)

You MUST document failures. Describe what failed, why you think it failed, test at least two fixes, and evaluate which fix worked and why.

1. Do all heads collapse to the same attention pattern? Hypothesize and test why.
2. Ablate to 1 head: does performance degrade? By how much and why?
3. Test random head dropout during training. Hypothesize the effect before running it.
4. What is each head actually attending to? Can you predict a head's role from its weights?

Mathematical Integration

1. Compute the effective rank of W_Q , W_K , W_V for each head at convergence
2. Analyze whether lower-rank weight matrices correlate with head specialization
3. Derive (on paper) why multi-head attention can represent functions single-head cannot
4. Measure the Frobenius norm of the W_O projection -- does it bottleneck information?

Research Questions to Answer

- What is each head learning? Can you assign a semantic role to each head?
- Can you surgically ablate one head and correctly predict what output behavior breaks?
- Does increasing head count always improve performance, or is there diminishing returns?
- What is the relationship between embedding dim, head count, and head dim?
- Does MSE vs cross-entropy change which heads form and how quickly?

Phase Progression

Phase	Focus
Phase 1	Build the core architecture
Phase 2	Compare with baseline / non-attention models

Phase 3	Study optimization dynamics
Phase 4	Modify loss functions and evaluate
Phase 5	Alter architecture systematically
Phase 6	Design novel training techniques

Document everything like a research paper. Your goal: independently analyze neural architectures, diagnose training failures, and propose novel improvements grounded in theory and empirical evidence.